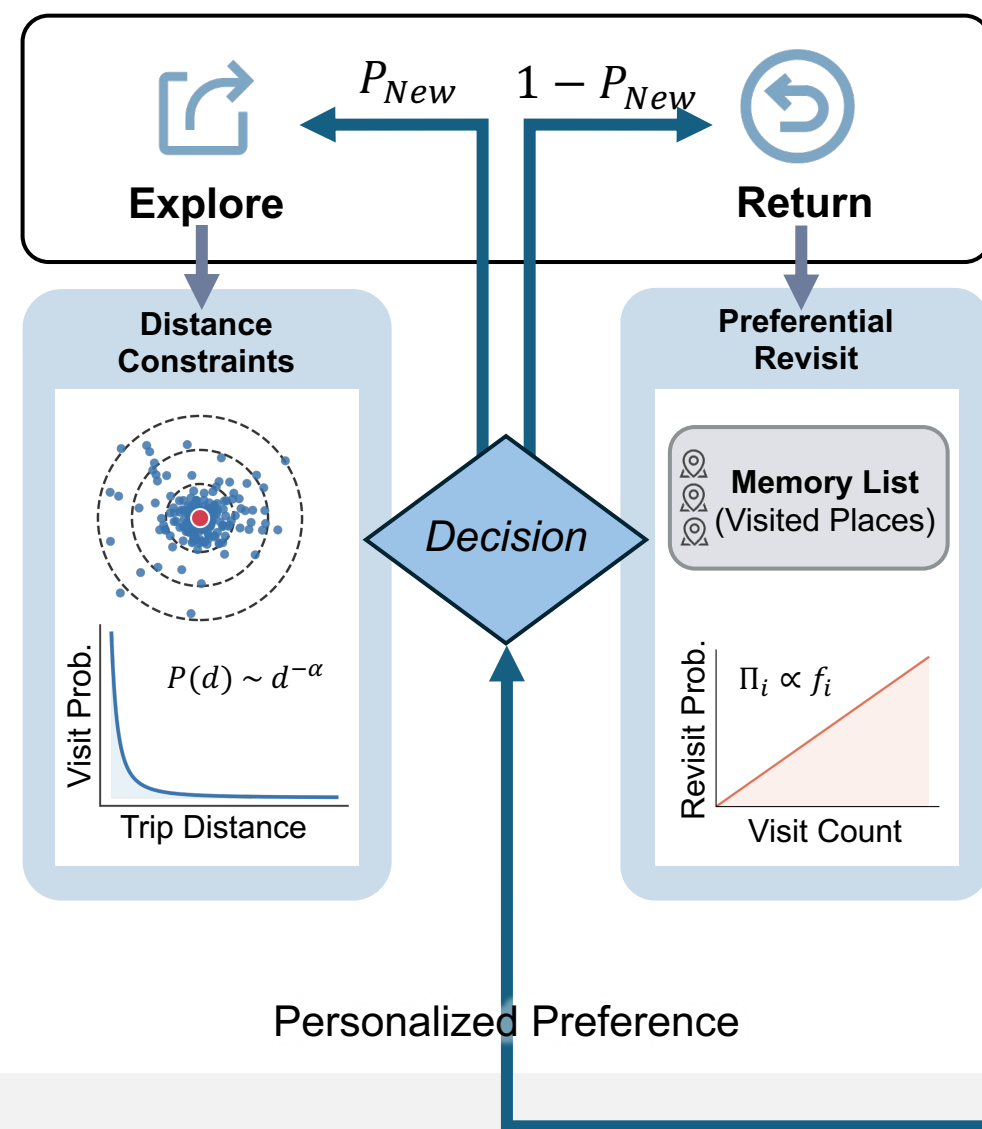
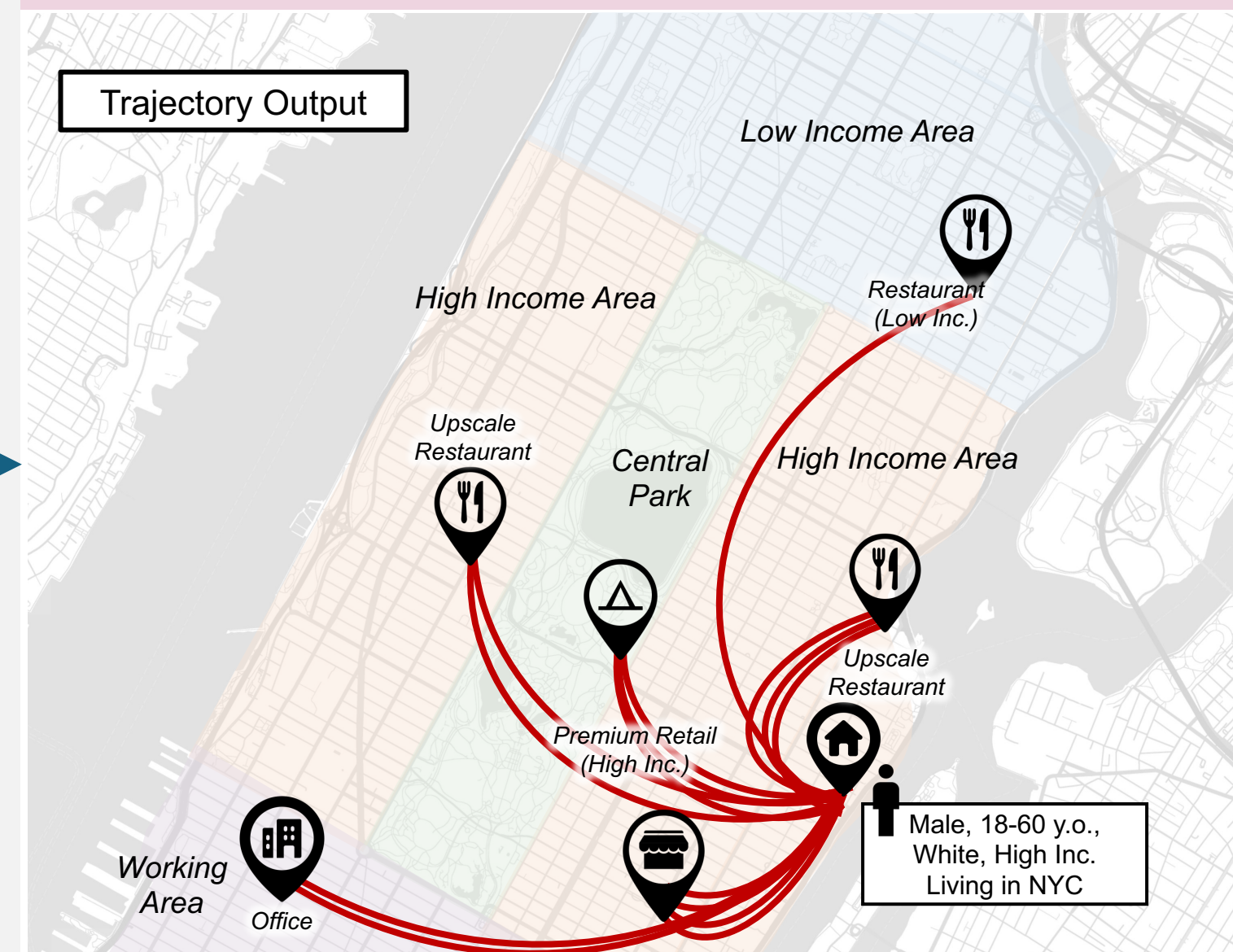


### (a) Population Level: Theory Input (EPR)

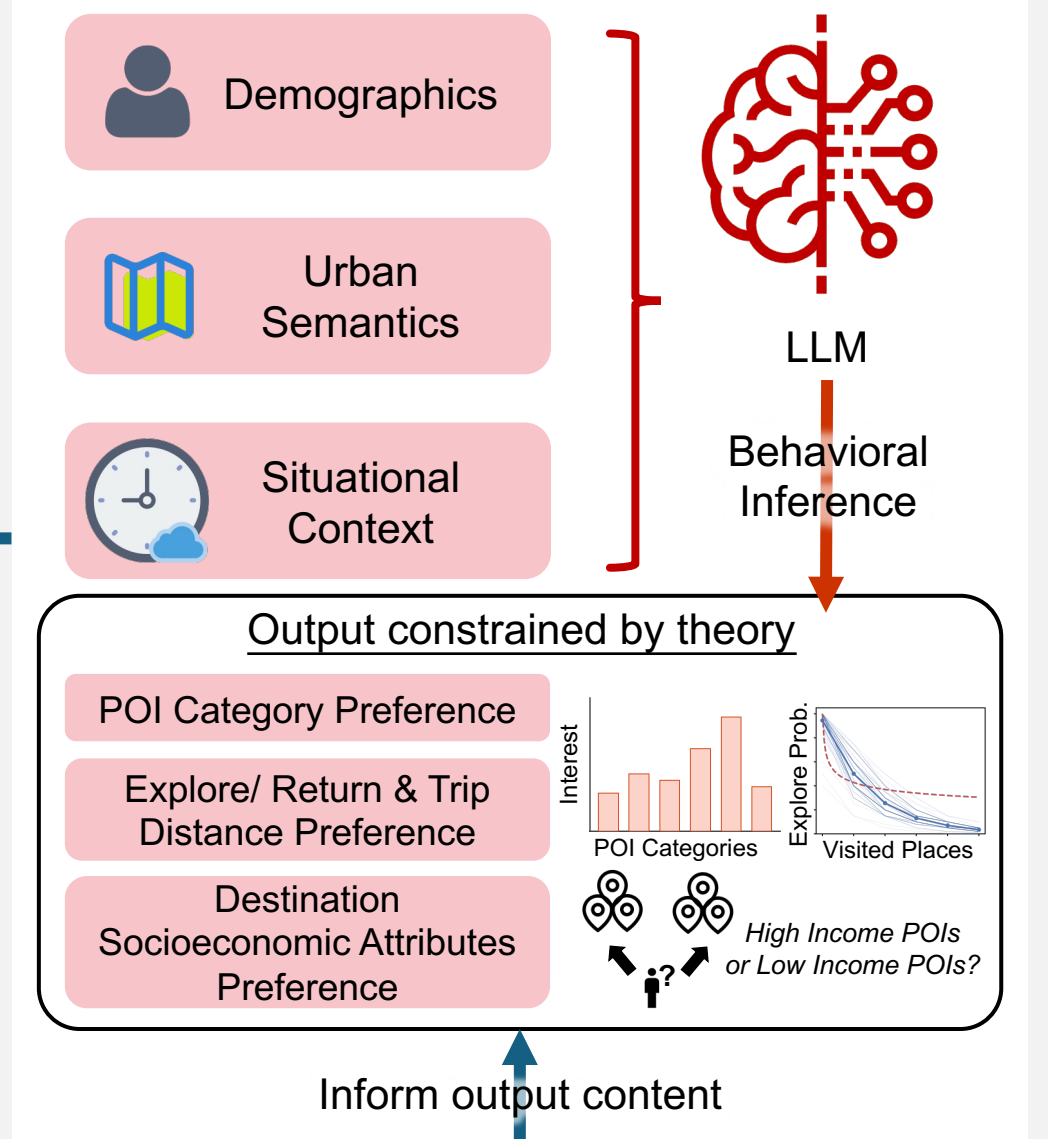
Decision using LLM-inferred behavior priors



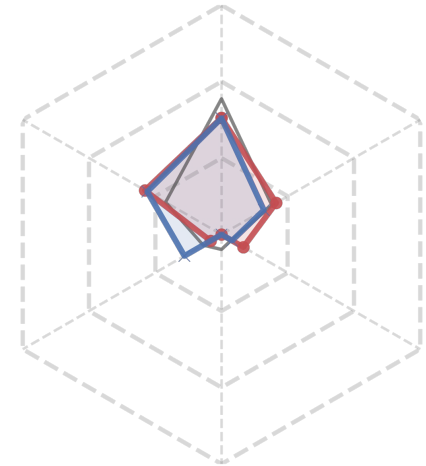
### (c) TIMA Framework



### (b) Profile Level: LLM-based Behavioral Prior



#### Micro-level Heterogeneity

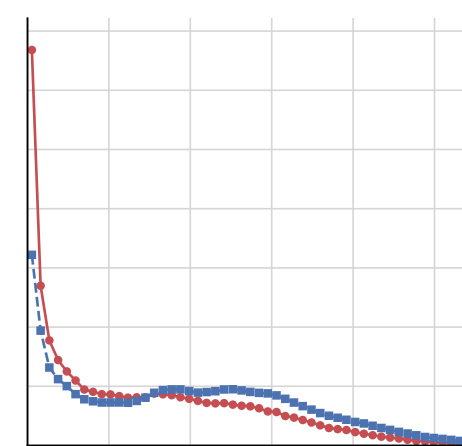


Personalized Mobility Pattern

TIMA agents exhibit diverse lifestyles conditioned on demographics and context

— City Avg  
—●— Real  
—×— TIMA

#### Macro-regularity Alignment



Overall Trip Distance

TIMA recovers the heavy-tailed nature of human travel without training on mobility data

—●— Real Data  
—×— TIMA

#### Emergent Social Segregation

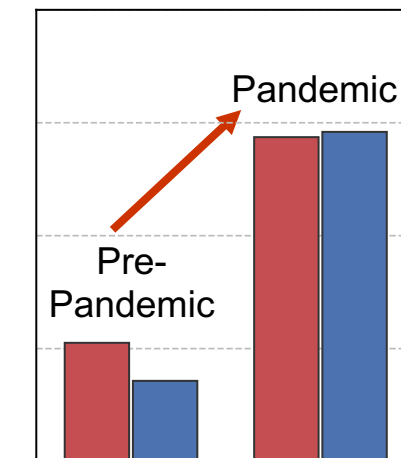


Social Segregation Map

Individual preferences accumulate into city-scale socio-economic stratification

Low High

#### External Crisis Adaptation



Home-stay Rate During Pandemic

LLM-based behavioral inference enables profile-specific mobility responses to pandemic shocks.

— Real Data  
— TIMA

### (d) TIMA Outcomes